

Development of a RASSOR Rover Remote Operation System with AI-Driven Digital Twin for
Latency Compensation

TO-BE Report

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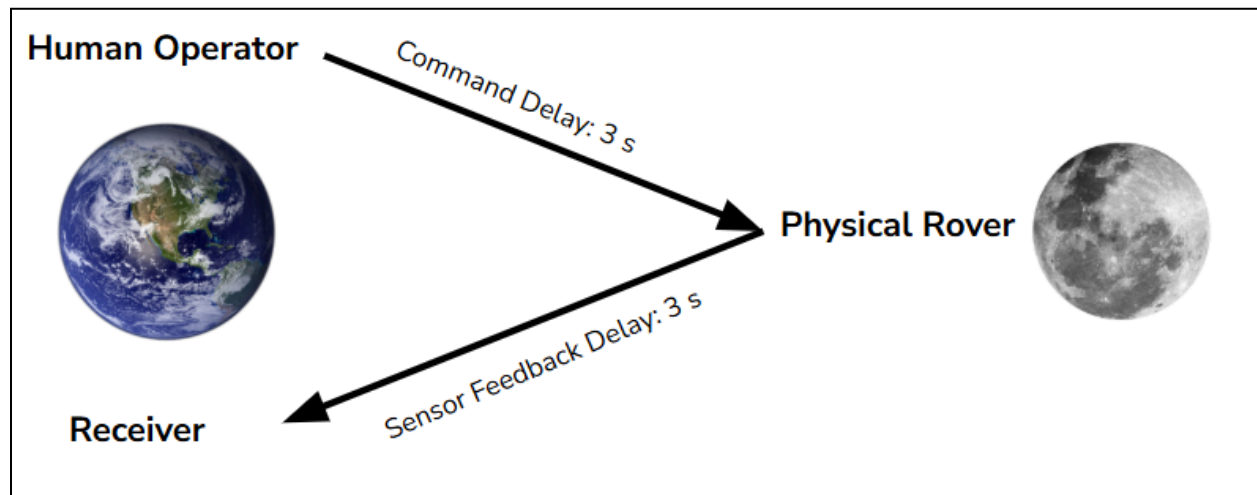
1. DMADV - Define:

1.1 Process and Problem Definition

The process under examination is the remote teleoperation of planetary rovers under communication latency conditions, specifically within Earth-to-Moon mission scenarios. Traditional teleoperation systems rely on near real-time feedback, however, lunar operations introduce an inherent ~3-second one-way communication delay, resulting in a ~6-second round-trip delay between command execution and feedback reception.

This delay fundamentally alters the human-rover control loop, creating a system where operators act on outdated information. As a result, the current process is characterized by:

- Reduced situational awareness
- Delayed hazard response
- Increased navigation errors
- Elevated risk of rover damage and mission inefficiency



The need for improvement lies in transitioning from a reactive teleoperation model to a predictive-assisted control system capable of compensating for latency effects. This project proposes the integration of a predictive digital twin framework with RGB-D sensing into a RASSOR rover system to address this limitation.

1.2 Problem Statement

Remote operation of lunar rovers is constrained primarily by communication latency, not mechanical capability. The ~3-second one-way delay (6-second round-trip) introduces a measurable performance gap in control accuracy and system responsiveness.

Key contributors include:

- Physical limits of long-distance signal transmission
- Teleoperation systems not designed for delayed feedback
- Network variability (latency fluctuations, packet loss, jitter)
- Human cognitive limitations under delayed feedback conditions

These factors result in:

- Increased obstacle collision frequency
- Higher path deviation from intended routes

- Longer task completion times
- Reduced operational safety and mission reliability

This project addresses a quantifiable performance gap between:

- Manual teleoperation under latency, and
- Predictive-assisted teleoperation using a digital twin system

1.3 Goals and Objectives

The primary goal is to improve rover teleoperation performance under simulated Earth-to-Moon latency conditions through predictive modeling and enhanced sensing.

Objectives:

- Demonstrate reliable rover operation under ~3-second simulated latency
- Reduce obstacle collision frequency across three modes:
 - Manual-only operation
 - Digital twin-assisted operation
 - AI-implemented predictive operation
- Reduce average obstacle course completion time under latency
- Reduce path deviation from a predefined optimal route
- Improve operator situational awareness and decision accuracy

These objectives align with stakeholder goals, Florida Space Institute, of improving safety, reliability, and operational efficiency in latency-constrained environments.

1.4 Performance Measures

Key Process Output Variables (KPOVs) used to evaluate system performance include:

- Obstacle Collision Frequency (count per trial)
- Course Completion Time (seconds)
- Path Deviation (distance from optimal path)
- Operator Response Delay (time between hazard appearance and corrective action)
- System Reliability (successful task completion rate)
- Prediction Accuracy (closeness of digital twin simulation to actual movements)

These metrics directly reflect system quality, efficiency, and reliability under latency conditions.

1.5 Scope Statement

In Scope

- Construction of a functional RASSOR rover platform
- Implementation of internet-based teleoperation
- Simulation of communication latency (approximately 3 seconds one-way)
- Integration of RGB-D sensing
- Development of a predictive digital twin framework
- Experimental testing and performance evaluation
- Documentation of system architecture and results

Out of Scope

- Deployment in actual lunar or space environments
- Use of space-qualified hardware

- Development of satellite communication systems
- Full autonomous rover replacement
- Long-term durability or environmental stress testing

The project focuses on proof-of-concept validation within controlled environments.

1.6 Needs, Requirements, and Constraints

Technical Requirements

- Real-time RGB-D data acquisition
- Accurate rover state prediction during latency
- Controlled latency injection in communication system
- Functional teleoperation user interface

Operational Requirements

- Usability for human operators
- Stability under simulated network conditions
- Repeatable testing procedures

Constraints

Technical Constraints

- Sensor accuracy and calibration limitations
- Prediction model uncertainty
- Hardware integration limitations

Economic Constraints

- Limited budget for hardware and computing resources

Environmental Constraints

- Testing limited to indoor or controlled outdoor environments

Regulatory Constraints

- No direct compliance with space-grade system standards required

Safety Constraints

- Prevention of rover damage during testing
- Safe operation within controlled environments

1.7 Engineering Design Considerations

The system design must balance performance improvement with reliability and usability under latency conditions.

Key Considerations

- Tradeoff between prediction accuracy and computational cost
- Human reliance on predictive visualization vs real sensor data
- Robustness to network variability and data inconsistencies
- Integration of sensing, modeling, and control systems

Risks and Limitations

- Prediction inaccuracies in the digital twin model
- Operator overreliance on predicted states
- Sensor calibration errors affecting mapping accuracy

- Mismatch between simulated and real-world latency conditions

2. Measure Phase - Current Process State

The measure phase focuses on the current performance of the rover under the simulated Earth-to-Moon communication delay and checking to make sure that the simulation matches the real life rover movements from the distance it moves to the conditions the rover faces from what is recorded through the RGB-D camera. Using the RGB-D camera to provide a layout of the environment the rover is in, we can synchronize the environment to the digital twin to simulate the rover there and all potential obstacles it encounters.

The Key Performance Input Variables (KPIVs) are the communication delay, rover speed, obstacle density, sensor processing capability, synchronization accuracy from the physical rover to the digital twin, and the data from the RGB-D. The Key Performance Output Variables (KPOVs) are the frequency of collisions, task completion time, path deviation, and operator situational awareness.

The synchronization between the physical rover and digital twin was addressed first to ensure both systems moved in unison. Accurate synchronization was essential before measuring the other KPIVs, as any mismatch would distort all subsequent data collection. To calibrate the movement, the distance covered by one wheel rotation on the physical rover was measured and then replicated in the digital twin. This was repeated 20 times to ensure consistency and precision. Furthermore we needed to make sure that the rover will input the commands the same so we ran multiple trials of giving it various combinations of inputs to ensure that the physical rover and digital twin match up perfectly. The goal was to have a less than 2 cm synchronization error measured through frame-by frame comparison of the position data between the physical rover and the digital twin.

The rover must maintain a stable teleoperation under the delay latency of 3 seconds as the digital twin will move first and then the rover will try to match the movement that was inputted by the user. Using a timer we made sure that the physical rover will move 3 seconds after the digital twin rover moves. Moreover we further measured it through time-delay analysis under communication logs.

The rover speed was measured through calculating the speed through the rotations times the wheel circumference divided by the elapsed time found through the RGB-D sensor log and physical timing. It is then compared between the rover's physical position updates with the digital twin's simulated position. This was done at different levels of speed with measuring each 20 times to ensure that no matter the level of setting the rover is running at the distance and speed are accurate between the physical version and digital version.

The data from the RGB-D showed promise as we were able to show some visuals from the camera from the rover to the laptop. We were able to have the rover distinguish certain objects placed in front of it, but the processing took a significant amount of time to run. Currently it will update as objects move around the camera, but when the rover itself moves around it needs to be a slow input level otherwise the data shown has notable gaps in the visualization.

The target performance for the obstacle avoidance is to have zero collisions. We are still in the process of getting to the standard of full visualization of the environment in order to test the rover on the amount of collisions and also with the path deviation. The goal for the path deviation is to be less than 5%.

Overall, we were able to collect quantitative data of the rover's synchronization accuracy between the physical rover and digital twin, qualitative data of the RGB-D data of the environment, and quantifiably measure the rover's speed in both the physical rover and digital twin to ensure synchronization between the two.

3. Analyze Phase - Performance Gaps

The Analyze phase focuses on finding the input variables that most affect rover performance under simulated Earth-to-Moon communication delay and evaluating how the implemented system improves rover operation under those conditions. Since the system now includes synchronized telemetry, RGB-D sensing, and a digital twin framework, the analysis examines how these additions improve operator awareness and reduce uncertainty during delayed teleoperation.

The main Key Performance Input Variables (KPIVs) are communication delay, rover speed, obstacle density, sensor processing capability, synchronization accuracy between the physical rover and digital twin, and RGB-D environmental data quality. The main Key Performance Output Variables (KPOVs) are collision frequency, task completion time, path deviation from the intended route, and operator situational awareness. Among these inputs, communication delay remains the most critical because operators must still make decisions under delayed feedback conditions. However, the implementation of the digital twin framework reduces the mismatch between the rover's actual state and the operator's perceived state during teleoperation.

This analysis shows that the main performance gap in rover operation is the disruption in the human-rover control loop caused by delayed state awareness. As the delay increases, the rover continues moving before the operator can react to updated feedback, increasing navigation error, repeated corrective actions, and collision risk. These effects become more severe as rover speed increases or obstacle environments become more complex. The integration of RGB-D sensing and synchronized telemetry improves the system's ability to capture environmental and

positional data, allowing the digital twin to provide a more accurate representation of rover behavior during communication delay.

The Analyze phase also evaluated the effectiveness of the communication and sensing systems. Using the ROS 2 topic-based communication structure, both the physical rover and digital twin receive identical command and telemetry streams. This allows the digital twin to mirror rover movement within the Isaac Sim environment while the physical rover executes delayed commands. As a result, the system improves synchronization between the physical and virtual rover systems and establishes the foundation for predictive modeling and future AI-assisted navigation.

The integration of the OAK-D Lite RGB-D camera enabled successful acquisition and visualization of point cloud data within the system environment. While the project did not progress to full environmental reconstruction or mapping integration, testing confirmed that the sensor could provide usable spatial and depth data for future digital twin development. The analysis showed that sensor performance and data quality were strongly influenced by processing capability and system throughput. System upgrades, including the transition from the Le Potato single-board computer to a Raspberry Pi, improved overall sensor stability, data handling, and point cloud visualization performance.

The analysis also identified power management as a remaining operational limitation. Simultaneous rover mobility, sensor streaming, and RGB-D data acquisition create high power demand, reducing runtime during testing. The camera system was identified as the primary contributor to battery drain, creating a tradeoff between environmental data quality and operational duration.

Three operational approaches were evaluated during the analysis process. Enhanced manual teleoperation, predictive digital twin-assisted teleoperation, and future AI-enhanced autonomous navigation. Manual teleoperation improves environmental awareness but still relies on delayed feedback. AI-enhanced navigation offers greater long-term autonomy but requires additional development and model validation. The predictive digital twin-assisted approach provides the most balanced solution for the team currently because it directly addresses the loss of situational awareness caused by communication latency while maintaining the human operator within the control loop. Based on this analysis, the implemented framework using synchronized telemetry, ROS 2 communication, RGB-D sensing, and the Isaac Sim environment provides the strongest foundation for reducing uncertainty during delayed rover operation and supporting future AI-assisted navigation development.

4. Design Phase (DMADV) - Process and System Design Alternatives

The Design phase focuses on developing and evaluating system-level solutions for rover operation in latency constrained environments, such as lunar exploration. Unlike traditional systems that attempt to eliminate delay, this project is designed to maintain performance and

situational awareness despite unavoidable communication latency. Consistent with ABET's definition of engineering design, this phase applies an iterative and decision-driven process that balances system performance with technical constraints, cost, and future scalability.

This project is intentionally structured as Phase 1 of a multi-phase system, where the primary objective is to develop a functional digital twin framework, rover platform, and data pipeline. This foundation enables Phase 2 development of AI-assisted navigation and autonomy.

4.1 Alternative Generation

Based on the Analyze phase, where latency was identified as a critical constraint in remote operations, three design alternatives were developed:

- Enhanced Manual Teleoperation with Sensor-Augmented Feedback
- Synchronized Digital Twin System (Selected Phase 1 Design)
- AI-Enhanced Digital Twin with Autonomous Navigation (Future Phase 2)

In addition to system-level alternatives, design decisions were also made at the hardware level, including rover structural modifications and sensor selection to support environmental mapping and future system expansion.

4.2 Conceptual Design of Alternatives

Rover Design and Hardware Modifications

The physical rover platform was developed from previous RASSOR models, with modifications to improve functionality, integration, and scalability. A key design change involved extending the shorter side of the chassis to match the longer side, creating a square geometry.

This adjustment increased internal space for wiring, electronics, and sensor integration, improving system organization and maintainability. The design supports modular expansion, allowing additional sensors and computational components to be integrated in future phases.

Sensor Selection and Environmental Mapping

Sensor selection was guided by performance requirements and cost constraints. High-performance LIDAR systems were initially considered due to their accuracy in mapping and navigation; however, these systems were determined to be cost-prohibitive. An RGB-D camera, OAK-D Lite, was selected as a cost-effective alternative. This sensor provides depth perception and generates point cloud data, enabling environmental mapping and obstacle detection.

The RGB-D system supports:

- Real-time spatial data acquisition
- Point cloud generation for terrain and obstacle mapping
- Data collection for future AI model training

This decision balances functionality and affordability while ensuring compatibility with future autonomous capabilities.

Enhanced Manual Teleoperation with Sensor-Augmented Feedback

This approach enhances the baseline system by incorporating RGB-D sensor data into the operator interface. The operator receives improved environmental visualization, including depth and obstacle information.

However, the system remains fully dependent on delayed feedback, limiting its effectiveness in latency-constrained environments. While perception is improved, the control strategy remains reactive, and latency effects are not mitigated.

Real-Time Digital Twin-Assisted Teleoperation (Phase 1 Design)

The selected design integrates a real-time digital twin developed in NVIDIA Isaac Sim, synchronized with rover telemetry and RGB-D sensor data. This system is designed for operation in environments where communication delays are unavoidable.

The system operates by receiving delayed sensor and telemetry data (simulated ~approx. 3-second delay), updating a virtual rover model within the simulation environment and providing a structured visualization of the rover and its environment using point cloud data, rather than eliminating latency, this approach improves operator performance by presenting organized and interpretable system information under delayed conditions. Controlled delays are introduced during testing to replicate real-world communication constraints.

The system establishes a data collection and mapping pipeline, enabling future development of predictive models and autonomous navigation. This Phase 1 implementation focuses on building the infrastructure required for Phase 2 capabilities.

AI-Enhanced Digital Twin with Autonomous Navigation (Phase 2 Vision)

This future design builds upon the digital twin framework by incorporating AI-driven navigation and decision-making. Using environmental and operational data collected in Phase 1, machine learning models can be trained to generate optimized navigation paths and perform obstacle avoidance.

In this system, the operator transitions from direct control to supervisory control, issuing high-level commands while the system executes navigation tasks. With this approach, it offers the highest performance potential, for quality and durability, it is beyond the scope of the current project due to its complexity and reliance on a fully developed data infrastructure.

4.3 Trade-Off and Feasibility Analysis

A weighted decision matrix was used to evaluate the alternatives based on key engineering and managerial criteria, including latency mitigation, implementation feasibility, system complexity, cost, computational demand, and scalability.

Criteria	Weight	Manual	Digital Twin	AI System
Latency Mitigation	0.30	2	4	5

Implementation Feasibility	0.20	5	4	2
System Complexity	0.15	5	3	1
Cost	0.10	5	3	2
Computational Demand	0.10	5	3	2
Scalability	0.15	2	4	5
Total Score	1.00	3.55	3.75	3.35

The real-time digital twin achieved the highest overall score by balancing performance improvement with implementation feasibility. While the AI-based system offers greater long-term capability, it introduces significant complexity and development risk. The manual system, although simple, does not adequately address the challenges of latency-constrained operation.

4.4 Engineering Constraints and Design Considerations

The design process was guided by several constraints:

- Fixed communication latency in space applications
- Hardware limitations of the rover platform and RGB-D sensor
- Computational requirements for simulation and data processing
- Data processing capacity constraints with Le Potato
- Energy constraints related to rover battery capacity
- Integration with ROS and NVIDIA Isaac Sim environments

These constraints informed the selection of a scalable and feasible design that minimizes reliance on real-time feedback.

4.5 Iterative Refinement and System Improvement

An iterative design process was implemented through system integration and testing. During development, an unintended delay in the control system was identified.

Through hardware isolation and signal path analysis, the issue was traced to a software loop within the command structure. The control system was refactored to eliminate this delay, resulting in near-instantaneous response under no-delay conditions. A controlled delay was then introduced to simulate realistic communication latency.

Final system testing demonstrated successful:

- Remote rover control
- Digital twin synchronization
- Environmental mapping using RGB-D point cloud data

4.6 Performance Evaluation

The system was evaluated against the baseline teleoperation approach to assess performance under simulated latency conditions. Results indicate measurable improvements in operator situational awareness, reduced navigation uncertainty, and more structured environmental understanding. By organizing sensor and system data through the digital twin interface, the operator is better equipped to make decisions despite delayed feedback. These improvements demonstrate that the proposed design more efficiently supports operation in latency-constrained environments compared to the baseline system.

4.7 Final Design Synthesis

The real-time digital twin-assisted system is selected as the final design for Phase 1. This system satisfies all established requirements by enabling effective rover operation in latency-constrained environments while remaining feasible within current technical and resource limitations. The final system includes a modified RASSOR rover platform with improved hardware integration with an RGB-D sensor-based environmental mapping system and a synchronized digital twin capable of remote operation and data collection. This design establishes a scalable foundation for Phase 2, where AI-driven navigation and autonomy can be developed using collected data. The system is prepared for future validation in simulated lunar environments, supporting applications in space exploration and other latency-sensitive operations.

5. DMADV - Verify: Final Validation, Transfer, and Continuous Improvement

The Verify Phase confirmed that the final system met the main Phase 1 project goals and was ready for future development. Verification focused on the rover prototype, RGB-D camera system, simulated communication delay, digital twin framework, and hand-off plan. Since this project is the first phase of a larger AI-assisted rover system, the goal was not full autonomy yet, but to prove that the hardware, sensing, and simulation foundation could work together.

The final design was evaluated based on the project goal of improving remote rover operation under simulated Earth-to-Moon communication latency. The completed system includes a functional RASSOR-style rover, camera-assisted teleoperation, RGB-D data collection, and a digital twin model in NVIDIA Isaac Sim. The rover responded to basic movement commands such as forward, backward, left, right, and stop. The digital model was also prepared by importing the CAD design into Isaac Sim and setting up the joints, colliders, and simulation environment. This confirmed that the system met the Phase 1 requirement of creating a working platform for future predictive control.

Prototype testing included controlled rover driving, camera testing, delay testing, and digital twin setup. One major test involved communication delay. Early testing showed that the delay was inconsistent and higher than expected. After troubleshooting, the team found that an

extra software loop was adding unintended delay. Once this was corrected, the rover operated with the intended 3-second command delay, helping verify that the system could represent the latency issue the project was designed to address.

The RGB-D camera system was also tested during verification. The camera produced live point cloud data, showing that it could collect usable depth and spatial information from the rover's environment. This data can support future digital mapping, obstacle detection, and digital twin development. Although the digital twin is not fully predictive yet, the simulation framework was created and can be expanded in later phases.

The main verification measures included command delay, sensor functionality, rover reliability, battery life, and readiness for continued digital twin development. The most important result was achieving the intentional 3-second delay. The camera system also met its basic goal by collecting usable point cloud data. However, battery life remains a major limitation. The rover currently runs for about 1 hour and takes about 3 hours to recharge, mainly because the camera system and rover movement draw a large amount of power at the same time.

Documentation and technology transfer are important because this project is expected to continue beyond the current team. The hand-off should include the rover hardware setup, software setup, sensor configuration, testing procedures, digital twin files, and notes on major issues found during development. Future teams should also receive basic operating procedures for starting the rover, running the camera, charging batteries, testing the delay, and using the Isaac Sim model. Version control should be used for software files, CAD models, simulation files, testing data, and documentation so that future changes can be tracked clearly.

Stakeholder and user feedback helped guide the final system. The Florida Space Institute provided support with electrical setup, Arduino configuration, battery integration, and troubleshooting. Additional feedback from the FSI rover group helped the team make better decisions about rover assembly, stability, and component placement. Testing feedback also led to improvements, including identifying the software loop that caused the delay issue. Battery life was also identified as a major concern, leading to the recommendation of a dual-battery system.

The recommended implementation and operating plan is to use a repeatable testing process each time the rover is operated. Before testing, future teams should inspect the rover hardware, check battery charge, confirm the camera connection, open the control software, and prepare the Isaac Sim digital twin files. During testing, the rover should first be driven without delay to confirm normal movement, then tested with the intentional 3-second delay to evaluate latency performance. The team should record command delay, rover response, battery life, camera performance, and any hardware or software issues after each trial. After testing, batteries

should be recharged, data should be saved, documentation should be updated, and any design changes should be tracked through version control.

The final system should be treated as a Phase 1 prototype that future teams can improve. The main areas for continued development are power management, onboard computing, sensor processing, digital twin accuracy, expanded testing, and AI integration. A dual-battery system is recommended so that the camera and motors do not rely on the same power source. Future testing should also include controlled obstacle courses and eventually sandy or regolith-like environments.

Overall, the Verify Phase showed that the team successfully built and tested a working rover prototype, validated RGB-D camera data collection, achieved the intended 3-second latency simulation, created a usable digital twin framework, and prepared the system for future development. While more work is needed before full AI-assisted rover control is possible, the current system provides a strong foundation for the next phase of the project.

6. Summary, Conclusions, and Future Work

6.1 Summary and Lessons Learned

The application of the Lean Six Sigma DMADV methodology provided a structured approach for the development of the latency-resilient rover system. Each phase contributed to refining the system design under the constraints of delayed communication. The Define phase established the operational problem and system objectives, while the Measure and Analyze phases enabled the collection and evaluation of performance data under simulated latency conditions. These results informed the Design phase, where multiple control strategies were considered, leading to the selection of a predictive digital twin-assisted approach.

Mathematical modeling and engineering analysis were used to support system development and decision-making. Kinematic relationships were applied to estimate rover motion during communication delays, and performance metrics such as response latency, path deviation, and task completion time were considered to evaluate system effectiveness. Data collected from testing environments provided a quantitative basis for comparing design alternatives and selecting the most feasible solution.

Several key lessons were identified throughout the project. Hardware integration required more time and coordination than initially anticipated, particularly in aligning sensor systems, onboard processing, and communication interfaces. The strength of the camera's processing demanded additional power supply that our initial designs did not account for. Trade-offs between system accuracy and computational efficiency were consistently observed, especially when considering predictive modeling approaches. Additionally, the project emphasized the importance of designing with latency as a primary constraint rather than treating it as a secondary limitation.

The iterative design process enabled the team to balance system requirements, constraints, and risks. For example, increasing sensor resolution improved environmental awareness but introduced additional processing delays. Similarly, higher rover speeds improved efficiency but increased sensitivity to communication latency, reinforcing the need for predictive compensation. Adjustments to project scope and scheduling also highlighted the importance of flexibility in managing engineering projects.

The current system is based on several assumptions, including controlled testing environments and simplified terrain conditions. These assumptions limit the generalizability of results to more complex or dynamic environments. Future work will need to address these limitations through expanded testing and improved environmental modeling.

6.2 Risks and Future Work

Several risks remain associated with system scaling and deployment. The accuracy and reliability of the predictive digital twin under varying terrain conditions present a primary concern. Inaccurate predictions may reduce operator effectiveness and increase the likelihood of navigation errors. System integration also introduces risk, particularly in maintaining synchronization between the physical rover and the digital twin under variable communication delays. Additionally, increased computational requirements for advanced predictive models may exceed current onboard processing capabilities.

Future work will focus on advancing the system through integration of an AI-driven predictive model within the digital twin framework. This model will utilize real-time and historical sensor data to improve prediction accuracy and system responsiveness under latency conditions. Approaches such as machine learning-based sequence prediction or adaptive control methods may be explored to enhance system performance.

Further testing will be conducted under more complex and variable conditions, including unstructured terrain and varying delay intervals. These tests will provide additional data to evaluate system robustness and validate performance improvements. Hardware upgrades may also be considered to support increased computational and power demands associated with advanced predictive modeling.

Further development opportunities include improving the operator interface to better communicate predictive feedback and reduce cognitive load. Exploration of semi-autonomous control strategies may further enhance system performance by allowing the rover to perform low-level navigation tasks independently during communication delays. Continuous improvement will be supported through iterative testing, performance evaluation, and stakeholder feedback. These processes will ensure that future system refinements are guided by measurable outcomes and aligned with project objectives.

6.3 Design Integration and Professional Judgment

The project demonstrates the integration of engineering design, analytical methods, and data-driven decision-making in addressing the challenges of latency in remote robotic systems. The selection of a predictive digital twin-assisted approach reflects a balance between system performance, feasibility, and implementation constraints.

Engineering decisions were guided by quantitative analysis, system requirements, and practical limitations. The planned integration of AI-based predictive modeling represents a continuation of this approach, supporting further improvement in system capability. Overall, the project highlights the importance of structured design methodologies, iterative refinement, and informed decision-making in the development of complex engineering systems.

7. Factors Considered and Their Importance to the Project (Design)

This section evaluates the key technical, operational, and ethical factors that the design of the latency-resilient rover system. The project required balancing performance, feasibility, cost, and long-term scalability while designing for operation in latency-constrained environments such as lunar missions. These considerations reflect the multidimensional nature of engineering design and its real-world constraints.

7.1 Comprehensive Consideration of Design Factors

Safety and Reliability (Including Human-System Interaction and Cybersecurity)

Safety and reliability are critical due to the remote nature of rover operation and the potential for system failure in hazardous environments. The system relies on human-in-the-loop control, making operator awareness and interface clarity essential to safe operation.

To address this, the design incorporates structured environmental visualization through the digital twin and RGB-D mapping, improving operator understanding under delayed feedback conditions. Reliability was further improved through iterative testing, including the identification and removal of unintended control delays in the command system.

Cybersecurity considerations include secure data transmission between the physical rover and digital twin, particularly important for future remote or mission-critical deployments. The trade-off, enhancing reliability and system feedback required increased system complexity and computational demand, particularly in maintaining synchronization between physical and virtual systems.

Economic Feasibility and Manufacturability

Cost constraints played a major role in design decisions. High-performance LIDAR systems were initially considered but ultimately rejected due to their high cost and limited accessibility. Instead, the team selected the OAK-D Lite RGB-D camera, which provides sufficient environmental mapping capability at a significantly lower cost. The rover itself was built using accessible components, including Arduino-based control systems and modular hardware, ensuring manufacturability and replicability for future teams. The trade-off, the lower-cost sensor provides less precision than LIDAR, but this was acceptable given the project's Phase 1 scope and emphasis on building a functional system foundation.

Regulatory and Legal Compliance

Although the project operates in a research and educational context, regulatory considerations remain relevant, particularly for future deployment. These include compliance with communication protocols, data handling practices, and safety standards for robotic systems. The use of standardized tools such as ROS and simulation environments like NVIDIA Isaac Sim supports compatibility with industry practices and future regulatory alignment. Designing within standardized frameworks may limit flexibility but improves interoperability and long-term applicability.

Functionality, Interoperability, and Scalability

Functionality was centered on enabling effective rover operation under latency conditions while supporting future system expansion. Interoperability was achieved through integration with ROS and Isaac Sim, allowing communication between the physical rover and digital twin. Scalability was a key design priority. The system was intentionally developed as Phase 1, focusing on building a data pipeline and infrastructure that can support future AI-driven navigation and autonomy. Prioritizing scalability required additional upfront design effort and system architecture planning, even though full AI functionality is not implemented in this phase.

Maintainability, Usability, and Ergonomics

Maintainability was addressed through rover design modifications, including extending the chassis to create additional space for wiring and electronic components. This improved accessibility and simplified future upgrades. Usability was improved through structured visualization in the digital twin, making environmental data more interpretable for the operator. Increasing system capability and visualization complexity may require additional operator training compared to simpler manual systems.

7.2 Ethical Analysis: Four Pillars of Reflexive Principlism

Respect for Autonomy

Respect for autonomy focuses on enabling users to make informed and meaningful decisions while maintaining control over system operation. In this project, autonomy is preserved through a human-in-the-loop control structure, where the operator remains responsible for navigation decisions rather than relying on full system autonomy. This is particularly important in latency-constrained environments, where delayed feedback can distort situational awareness. To support informed decision-making, the system integrates structured environmental visualization through the digital twin and RGB-D mapping, allowing the operator to better interpret surroundings despite communication delays. An ethical challenge arises in balancing automation with operator control. While increasing automation could improve efficiency, it may also reduce operator understanding and decision authority. To address this, the project intentionally limits automation in Phase 1 and prioritizes transparency and interpretability. Future AI integration (Phase 2) will require explainable decision-making to ensure that operators retain awareness and trust in system behavior.

Beneficence

Beneficence emphasizes designing systems that provide meaningful benefits to users, stakeholders, and society. This project contributes to safer and more efficient robotic operations in environments where direct human presence is not feasible, such as lunar exploration. By improving situational awareness under latency, the system reduces operational risk and enhances mission effectiveness. Beyond space applications, the system has potential benefits in Earth-based contexts, including hazardous mining, disaster response, and remote infrastructure inspection. In these scenarios, improved remote operation capabilities can reduce human exposure to dangerous conditions. Another aspect of beneficence is knowledge transfer. The project was intentionally designed as a foundational Phase 1 system, providing future teams with a working platform, data pipeline, and integration framework. This ensures that the work contributes to long-term advancement rather than a one-time implementation.

Nonmaleficence

Nonmaleficence focuses on identifying, preventing, and mitigating potential harm associated with system operation. In this project, several risks were considered, including rover collisions, operator error due to delayed feedback, system miscommunication, and hardware or software failure. A key ethical concern is that latency can cause operators to act on outdated information, increasing the likelihood of incorrect decisions. Rather than ignoring this constraint, the system was explicitly designed to operate under simulated delay conditions, allowing risks to be identified and addressed during development.

Mitigation strategies include structured visualization of environmental data to reduce operator confusion, controlled latency simulation to test system behavior under realistic conditions, iterative testing and debugging, including removal of unintended system delays, and designing a system that avoids over-reliance on automation in early stages

Another ethical consideration is the risk of overconfidence in system capabilities, particularly in future AI-assisted versions. By clearly defining the system as a Phase 1 framework, the team avoids overstating performance and ensures responsible development progression.

Justice

Justice focuses on the fair distribution of benefits, risks, and access to technology. In this project, fairness was considered in both system design and resource allocation. A key decision reflecting this principle was the selection of an RGB-D camera over a high-cost LIDAR system. While LiDAR offers higher precision, its cost creates barriers to accessibility and the ability of replicability. By choosing a more affordable sensor that still meets functional requirements, the project enables broader adoption and allows future teams to build upon the system without significant financial constraints, as well as offer potential cost options for the Florida Space Institute for future testing and space advancements

Justice also applies to the distribution of responsibility and risk. The decision to maintain human-in-the-loop control ensures that accountability remains clear and that critical decisions are not delegated to unvalidated autonomous systems. This is especially important in early-stage technologies where reliability is still being established. In future phases, considerations of justice will extend to ensuring that AI systems do not introduce bias or unequal performance across environments or use cases.

7.3 Integration into the Engineering Design Process

Technical, ethical, and operational factors were not treated as separate considerations but were continuously integrated throughout the design process, directly influencing system architecture, component selection, and overall project scope. Rather than optimizing for a single objective, the team balanced competing priorities such as performance, cost, safety, and scalability to develop a feasible and forward-compatible solution. Key design decisions reflect this integration. For example, the selection of the RGB-D camera over a LiDAR system was not solely a technical decision, but a trade-off between mapping capability, economic feasibility, energy consumption, and accessibility. While LiDAR offered higher precision, its cost and resource requirements conflicted with project constraints and long-term scalability. The RGB-D sensor provided sufficient functionality while enabling broader adoption and future development, aligning technical performance with economic and ethical considerations.

Similarly, modifications to the rover structure, specifically extending the chassis to create a square shape, were driven by maintainability and system integration needs. Increasing internal space improved wiring organization while also reducing the risk of hardware interference, and allowed for future sensor expansion, now potential an additional battery. This decision reflects the integration of usability, reliability, and scalability into the physical design, rather than focusing solely on immediate functionality.

The decision to structure the project as a two-phase system further demonstrates the integration of technical and strategic considerations. Instead of attempting to implement full autonomy within limited time and resources, the team prioritized developing a stable digital twin framework and data pipeline. This approach ensures that future AI capabilities can be built on a reliable foundation, reducing long-term risk and improving system sustainability. The iterative refinement process also highlights how real-world constraints shaped the design. During system testing, an unintended delay in the control architecture was identified, which conflicted with the goal of accurately simulating communication latency. Through debugging and system analysis, the team isolated the issue and restructured the control logic to eliminate the additional delay. A controlled delay was then intentionally introduced to replicate realistic operating conditions. This process demonstrates how testing, feedback, and constraint validation were used to improve system accuracy and reliability.

Ethical considerations were embedded in these decisions as well. Maintaining human-in-the-loop control in Phase 1 reflects a deliberate choice to preserve operator autonomy, ensure accountability, and avoid over-reliance on unvalidated autonomous systems. Additionally,

selecting accessible and cost-effective components supports fairness and enables future teams to build upon the system without significant resource barriers.

By integrating technical requirements with ethical, economic, and operational constraints, the final design is more robust, adaptable, and aligned with real-world engineering challenges. The system not only meets functional objectives but also supports long-term development, responsible implementation, and practical deployment in latency-constrained environments.

8. Sources of Knowledge, Standards, and Best Practices Consulted

8.1 Knowledge Foundations and Academic Integration

The development of this project did not rely on a single course or specific technical concept, but rather on the cumulative analytical and problem-solving skills developed throughout the engineering and data science curriculum. Foundational coursework provided the ability to approach complex problems using structured methodologies, critical thinking, and logical reasoning. Through prior academic experience, the team developed a systematic approach to engineering problem-solving, including the ability to decompose complex systems, analyze constraints, and evaluate potential solutions. These skills were essential in addressing challenges related to communication latency, rover control, and overall system design.

Although the project did not directly map to specific course material in its early stages, the methodologies and thought processes developed throughout the curriculum played a critical role in guiding decision-making. The ability to approach problems from an engineering perspective, emphasizing analysis, iteration, and optimization, enabled the team to develop effective and adaptable solutions. Additionally, industrial engineering methodologies, particularly the DMADV framework, guided the structured design and development process, ensuring that the system was developed with a focus on performance, reliability, and validation.

8.2 Expert Consultation and Collaborative Input

The project benefited from guidance and support provided by industry professionals and collaborative team members. Christian Vincent from the Florida Space Institute (FSI), who served as the project sponsor, played a significant role in the development process. He provided technical assistance with the electrical components of the rover, including support with Arduino setup, LePotato battery integration, and overall system configuration. His guidance was particularly valuable during the development and testing phases, where he assisted in troubleshooting issues and debugging system components to ensure proper functionality. Additional support was provided by Kevin, the group lead from FSI Group One, who contributed valuable insights based on his experience with the full manual assembly of his team's rover. His understanding of the mechanical design allowed him to offer practical recommendations to improve the stability and performance of our rover. This included guidance on component selection, such as appropriate hardware, as well as structural adjustments to enhance durability and operational reliability.

The input from both industry professionals and peer collaborators played a critical role in refining the system design, improving performance, and addressing technical challenges. Their feedback directly influenced key design decisions and contributed to the overall success and functionality of the project.

8.3 Reference and Knowledge Sources

A wide range of credible and relevant sources were consulted to support the development of this project. Key references included technical and research-based articles such as “*Network Latency: Understanding the Impact of Latency on Network Performance*” (Freedman, 2026) and “*Learn: Teleoperation*” (Linnell, 2024), which provided foundational insight into communication delays and remote robotic control systems. Additional research on digital twin technologies was informed by sources such as “*Digital Twin Technology Advancing Industry 4.0 and Industry 5.0 Across Sectors*” and “*A Review on Digital Twins for Smart Manufacturing*,” which contributed to the general understanding of predictive modeling and system simulation. Environmental and sustainability considerations were supported by sources such as “*Space Launches and the Environment*” (Donou-Adonsou, 2024) and “*How Space Exploration Benefits Life on Earth*” (Sierra Space, 2024), while broader economic and industry context was informed by NASA-related economic impact analyses. All sources were cited using APA format to ensure consistency, credibility, and academic integrity throughout the report.

8.4 Standards and Best Practices

The project incorporated several industry standards and best practices to ensure quality, consistency, safety, and ethical integrity throughout development. The DMADV methodology provided a structured framework that improved project organization, maintained focus, and ensured a systematic approach to design and validation. This contributed to consistent decision-making and overall system reliability.

Safety considerations were guided by established protocols, including Stanford Environmental Health & Safety guidelines for 3D printing. These standards ensured proper handling of materials, minimized health risks, and promoted safe testing and fabrication practices.

Ethical integrity was maintained through adherence to the ACM Code of Ethics and Professional Conduct, which emphasizes responsibility, reliability, and respect for professional standards in computing systems. The integration of these principles ensured that the system was developed with a strong emphasis on human-centered design, accountability, and responsible use of technology.

Collectively, these standards and best practices strengthened the overall quality of the project, improved consistency in development, enhanced operational safety, and ensured alignment with recognized ethical and professional guidelines.

9. Phase II, Recommendations, and Team Composition

9.1 Phase II Planning and Recommendations

This project was intentionally developed as Phase 1 of a larger long-term system focused on latency-aware robotic operation in lunar environments. The primary objective of Phase 1 was to establish a functional rover platform, digital twin framework, and environmental data pipeline that can support future development of AI-assisted navigation and autonomous capabilities.

While the current system successfully demonstrates synchronized remote operation and environmental mapping using RGB-D point cloud data, several opportunities remain for future expansion and validation.

The primary recommendation for Phase II is the implementation of AI-assisted navigation and state estimation capabilities. Using the environmental and telemetry data collected during Phase 1, future teams can train machine learning models for autonomous obstacle avoidance, path planning and optimization, rover state estimation under delayed communication conditions and adaptive navigation in changing terrain environments

Other recommendations are expanded testing and validation in simulated lunar environments such as the Exolith Lab. Testing in lunar analog terrain would allow future teams to evaluate rover mobility, mapping accuracy, and system reliability under more realistic operating conditions.

Future work should also focus on:

- Improving synchronization between the physical rover and digital twin
- Expanding sensor integration and data collection capabilities
- Enhancing system robustness under extended communication delays
- Optimizing power consumption and battery performance
- Refining operator interfaces for improved usability and situational awareness

Another improvement is implementing the communication methodology developed by the Polytechnic University of Durango to simulate realistic long-distance delay conditions. This would strengthen validation of the system's performance in latency-constrained environments.

Overall, Phase II should prioritize transitioning the system from a foundational digital twin framework toward semi-autonomous and eventually autonomous operation while maintaining reliable human oversight.

9.2 Team Composition and Interdisciplinary Collaboration

The project team consisted of seven students from interdisciplinary engineering and technical backgrounds:

- Jennifer Cifuentes (Industrial Engineering): Project Manager
- Abdiel Joseph (Industrial Engineering) : Risk and Process Engineer
- Noah Choi (Data Science): Assembly and Development
- Sean Crites (Data Science): Digital Model Specialist
- Matteo D'Agostino (Data Science): Assembly and Development
- Mattias Escudero (Data Science): Digital Twin and Systems Integration
- Jonathan Gyure (Data Science): Data Pipeline Integration

The interdisciplinary composition of the team was essential to the project's success. Industrial Engineering students contributed project planning, workflow optimization, risk analysis, and systems-level decision-making, while Data Science students focused on simulation development, telemetry synchronization, data integration, and digital twin implementation.

This collaboration enabled the project to address both technical and operational challenges simultaneously. The integration of multiple disciplines also reflects the nature of real-world engineering projects, where successful system development requires coordination between hardware, software, systems engineering, and process management perspectives.

Future phases of the project would benefit from additional interdisciplinary participation, particularly from mechanical engineering for rover mobility and structural optimization, electrical engineering for power systems and hardware integration, and computer science for autonomous navigation development, and possibly a human engineering for operator interface and usability improvements

The collaborative structure of the project strengthened communication, problem-solving, and overall system integration, providing a strong foundation for future continuation efforts.